

3.2 Properties of determinants

Theorem 3 Row operations

Let A be a square matrix.

- If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
- If two rows of A are interchanged to produce B , then $\det B = -\det A$.
- If one row of A is multiplied by k to produce B , then $\det B = k \det A$.

$R_2 \leftarrow R_2 + 2R_1$
 $R_3 \leftarrow R_3 + R_1$ } According to theorem 3.a, this doesn't change the determinant.

$$\begin{bmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{bmatrix}$$

$R_2 \leftrightarrow R_3 \rightarrow$ Row exchange, multiply determinant by (-1) .

$$\begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{bmatrix}$$

$$\det A = (-1)(1)(3)(-5) = 15$$

Example 1

Compute $\det A$, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$

Solution

The strategy is to use the fact that the determinant of a triangular matrix is the product of the main diagonal entries.

3.2 Properties of Determinants

II

A common use of theorem 3c in hand calculations is to factor out a common multiple of one row of a matrix. For instance,

$$\begin{vmatrix} * & * & * \\ 5k & -2k & 3k \\ * & * & * \end{vmatrix} = k \begin{vmatrix} * & * & * \\ 5 & -2 & 3 \\ * & * & * \end{vmatrix} \quad \text{where the star entries are unchanged}$$

Example 2

Compute $\det A$, where $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}$

To simplify arithmetic we want a 1 in upper left corner (a_{11}). Instead of swapping rows 1 and 4, which will induce a (-1) factor on the determinant, we will instead factor out a 2 from row 1.

$$\det A = 2 \begin{vmatrix} 1 & -4 & 3 & 4 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{vmatrix}$$

Perform row operations

$$R_2 \leftarrow R_2 - 3R_1$$

$$R_3 \leftarrow R_3 + 3R_1$$

$$R_4 \leftarrow R_4 - R_1$$

$$= 2 \begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & -12 & 10 & 10 \\ 0 & 0 & -3 & 2 \end{vmatrix} \quad \text{Here we choose to add four times row 2 to row 3 to get rid of the -12 entry}$$

$$R_3 \leftarrow R_3 + 4R_2$$

$$\det A = 2 \begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & -3 & 2 \end{vmatrix}$$

Last step is to add $-1/2$ row 3 to row 4

$$R_4 \leftarrow R_4 + (-1/2)R_3$$

$$\det A = 2 \begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$$= 2(1)(3)(-6)(1) = -36$$

Suppose a square matrix A has been reduced to an echelon form U by row operations. If there has been r interchanges, Theorem 3 shows that

$$\det A = (-1)^r \det U$$

3.2 Properties of Determinants (cont'd)

III

Theorem 4

A square matrix A is invertible if and only if $\det A \neq 0$.

$\det A$ will be zero if a column or row in A is zero, or if two columns or rows are equal.

Example 3

Compute $\det A$, where $A = \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$

Solution

$$R_3 \leftarrow R_3 + 2R_1$$

$$\begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{bmatrix}$$

Rows 2 and 3 are equal, and we can conclude that

$$\boxed{\det A = 0}$$

Example 4

This example uses a combination of row operations and the strategy from 3.1 of using zero entries in cofactor expansions.

Compute $\det A$, where $A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$

Solution

Let's start by adding R_2 to R_4

$\det A = \begin{vmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -3 & 1 \end{vmatrix}$ then use cofactor expansion by factoring out a 2 from R_2 (be aware of the sign)

$\det A = -2 \begin{vmatrix} 1 & 2 & -1 \\ 3 & 6 & 2 \\ 0 & -3 & 1 \end{vmatrix}$ next we subtract $3R_1$ from R_2

$\det A = -2 \begin{vmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & -3 & 1 \end{vmatrix}$ Here we can either swap R_2 and R_3 or make another cofactor expansion down R_1

Swap R_2 and $R_3 \Rightarrow \det A = (-2) \begin{vmatrix} 1 & 2 & -1 \\ 0 & -3 & 1 \\ 0 & 0 & 5 \end{vmatrix} = (-1)(-2)(1)(-3)(5) = \underline{\underline{-30}}$

Make cofactor expansion down R_1

$$\det A = -2 \begin{vmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & -3 & 1 \end{vmatrix} = -2(0 - (-15)) = \underline{\underline{-30}}$$

3.2 Properties of Determinants (cont'd)

IV

Theorem 5

If A is an $n \times n$ Matrix, then $\det A = \det A^T$

Theorem 6

If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$

Example 5

Verify theorem 6 for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

$$AB = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$$

$$\det A = 9, \det B = 5, \underline{\det A \cdot \det B = 45}$$

$$\underline{\det AB = 45}$$

In $\mathbb{R}^2$ $|\det [\vec{a} \ \vec{b}]|$ equals the area spanned by $\vec{a}$ and $\vec{b}$.

A linear property of the determinant function

For an $n \times n$ matrix A , we can consider $\det A$ as a function of the n column vectors in A . We will show that if all columns except one are held fixed, then $\det A$ is a linear function of that one (vector) variable.

Suppose that the j th column of A is allowed to vary, and write

$$A = [\vec{a}_1 \ \dots \ \vec{a}_{j-1} \ \vec{x} \ \vec{a}_{j+1} \ \dots \ \vec{a}_n]$$

Define a transformation T from $\mathbb{R}^n$ to $\mathbb{R}$ by

$$T(\vec{x}) = \det [\vec{a}_1 \ \dots \ \vec{a}_{j-1} \ \vec{x} \ \vec{a}_{j+1} \ \dots \ \vec{a}_n]$$

Then,

$$T(c\vec{x}) = cT(\vec{x}) \text{ for all scalars } c \text{ and all } \vec{x} \text{ in } \mathbb{R}^n$$

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v}) \text{ for all } \vec{u}, \vec{v} \text{ in } \mathbb{R}^n$$

3.2 Properties of Determinants (cont'd)

V

Example of det A's linear properties

Suppose we have a 2×2 matrix where we allow its second column to change.

$$\vec{a}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A = [\vec{a}_1 \quad \vec{x}]$$

Then suppose we define a transformation

$$T = \det A = \det \begin{bmatrix} 2 & x_1 \\ 3 & x_2 \end{bmatrix} = 2x_2 - 3x_1$$

this is a linear function (linear combination) of $\vec{x}$.

Let's use this setup to test the two linearity rules from page IV.

Let's define two vectors in $\mathbb{R}^2$, $\vec{u}$ and $\vec{v}$, and put them into $T()$ instead of $\vec{x}$.

$$\vec{u} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$T(\vec{u}) = \det \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} = 5$$

$$T(\vec{v}) = \det \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix} = 4$$

Let's add $\vec{u}$ and $\vec{v}$

$$T(\vec{u} + \vec{v}) = \det \begin{bmatrix} 2 & 1+2 \\ 3 & 4+5 \end{bmatrix} = 9$$

and indeed,

$$T(\vec{u}) + T(\vec{v}) = T(\vec{u} + \vec{v})$$

Let's test the second rule

$$T(3\vec{u}) = \det \begin{bmatrix} 2 & 3 \cdot 1 \\ 3 & 3 \cdot 4 \end{bmatrix} = 15$$

and we also see that

$$3T(\vec{u}) = 15$$



It is important to note that

$$\det(A+B) \neq \det(A) + \det(B)$$

The determinant is linear in ONE column at a time